



Practice Exercise 1

COMP1322 Programming II

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Array Manipulation & Reversal in Java

Code:

```
1 import java.util.Scanner;
2
3 public class ArrayManipulation {
4     public static void main(String[] args) {
5         Scanner scanner = new Scanner(System.in);
6
7         // input N with validation
8         int n = 0;
9         while (n < 1 || n > 10) {
10             System.out.print("Enter size (1-10): ");
11             n = scanner.nextInt();
12         }
13         scanner.nextLine(); // make sure to consume the newline after nextInt(), because first time program didn't run correctly due to this issue
14
15         // read N integers from one line
16         System.out.print("Enter " + n + " integers: ");
17         String[] strings = scanner.nextLine().trim().split(regex: "\\s+");
18         int[] A = new int[n];
19         for (int i = 0; i < n; i++) {
20             A[i] = Integer.parseInt(strings[i]);
21         }
22
23         System.out.println("Length of A=" + n);
24
25         // resize array
26         A = resizeArray(A, n);
27         System.out.println("Length of A=" + A.length);
28
29         reverseAndPrint(A, n); // reverse first N elements and print
30
31         scanner.close();
32     }
```

```
34 @ static int[] resizeArray(int[] A, int n) { 1个用法
35     int newSize = (n < 10) ? 10 : 15;
36     int[] newA = new int[newSize];
37     for (int i = 0; i < n; i++) {
38         newA[i] = A[i];
39     }
40     return newA;
41 }
42
43 static void reverseAndPrint(int[] A, int n) { 1个用法
44     for (int i = 0; i < n / 2; i++) {
45         int temp = A[i];
46         A[i] = A[n - 1 - i];
47         A[n - 1 - i] = temp;
48     }
49
50     System.out.print("Reversed Array: ");
51     for (int i = 0; i < n; i++) {
52         System.out.print(A[i]);
53         if (i < n - 1) System.out.print(",");
54     }
55     System.out.println();
56 }
57 }
```

Output:

```
Enter size (1-10): 5
Enter 5 integers: 1 2 3 4 5
Length of A=5
Length of A=10
Reversed Array: 5,4,3,2,1
```

```
Enter size (1-10): 10
Enter 10 integers: 10 20 30 40 50 60 70 80 90 100
Length of A=10
Length of A=15
Reversed Array: 100,90,80,70,60,50,40,30,20,10
```